Abhishek Bichhawat

AB 13/405A, IIT Gandhinagar
Gandhinagar-382355, India
✉ *abhishek.b@iitgn.ac.in*

Research Interests

Web security and privacy, Usable security and privacy, Verification of protocols and systems, Distributed systems and performance optimization

Professional Appointments

Jun. 2026 – Present **Associate Professor**, IIT Gandhinagar, Gandhinagar, India.
Department of Computer Science and Engineering

Mar. 2021 – Jun. 2026 **Assistant Professor**, IIT Gandhinagar, Gandhinagar, India.
Department of Computer Science and Engineering

Jan. 2018 – Mar. 2021 **Postdoctoral Fellow**, Carnegie Mellon University, Pittsburgh, USA.
School of Computer Science and Cylab

Aug. 2011 – Nov. 2012 **Member of Technical Staff**, Oracle India Pvt. Ltd., Bangalore, India.
ASM Team – Database Test Development

Apr. 2011 – Jun. 2011 **Visiting Scholar**, Macquarie University, Sydney, Australia.

Education

Jan. 2013 – Jun. 2018 **Ph.D. Computer Science**, *Universität des Saarlandes and IMPRS-CS*, Germany.
Thesis: Practical Dynamic Information Flow Control.
Advisors: Christian Hammer & Deepak Garg

Jul. 2009 – Jul. 2011 **M.Tech. Computer Science and Engineering**, *IIT Roorkee*, India.
First Class with Distinction - *CGPA – 9.61*

Aug. 2005 – Jun. 2009 **B.E. Computer Science and Engineering**, *Anna University, Chennai*, India.
First Class with Distinction - *85%*

Publications

ASPLOS 2027 **DigitsOnTurbo: Leveraging SIMD for Accelerating Large-number Arithmetic**

CCS 2026 **Usability Compliance vs. Legal Compliance of Consent Withdrawal on the Web: An Expert Evaluation of 200 Websites**

IICT 2026 **Complementary Function Co-scheduling Across Call Boundaries**

Oakland 2026 **"I Wonder if These Warnings Are Accurate": Security and Privacy Advice in Nine Majority World Countries**

FSTTCS 2025 **Fall-through Semantics for Mitigating Timing-based Side Channel Leaks**

PoPETs 2025 **Johnny Can't Revoke Consent Either: Measuring Compliance of Consent Revocation on the Web**

EASE 2025 **On the Prevalence and Usage of Commit Signing on GitHub: A Longitudinal and Cross-Domain Study**

WWW 2025 **Least Privilege Access for Persistent Storage Mechanisms in Web Browsers**

ICSE-NIER 2025 **A Unified Browser-Based Consent Management Framework**

ISEC 2025	**Worst-Case Response Time Analysis for Periodic Programs with Nested Locks**
BuildSec 2024	**Online Authentication Habits of Indian Users**
ICISS 2024	**Web Privacy Perceptions Amongst Indian Users**
CCS 2023	**Tainted Secure Multi-Execution to Restrict Attacker Influence**
ESORICS 2023	**Layered Symbolic Security Analysis in DY***
SOUPS 2023	**Towards Usable Security Analysis Tools for Trigger-Action Programming**
EuroSP 2022	**Compositional Information Flow Monitoring for Reactive Programs**
Oakland 2022	**Noise*: A Library of Verified High-Performance Secure Channel Protocol Implementations**
PSL 2022	**A Tutorial-Style Introduction to DY***
CCS 2021	**An In-Depth Symbolic Security Analysis of ACME**
JCS 2021	**Permissive Runtime Information Flow Control in the Presence of Exceptions**
EuroSP 2021	**DY*: A Modular Symbolic Verification Framework for Executable Cryptographic Protocol Code**
CSF 2021	**Gradual Security Types and Gradual Guarantees**
CSF 2021	**Automating Audit with Policy Inference**
AsiaCCS 2020	**Contextual and Granular Policy Enforcement in Database-backed Applications**
ESORICS 2017	**WebPol: Fine-grained Information Flow Policies for Web Browsers**
CSF 2015	**Information Flow Control for Event Handling and the DOM in Web Browsers**
ICSE 2015 Companion	**Post-dominator analysis for precisely handling implicit flows**
PLAS 2014	**Generalizing Permissive-Upgrade in Dynamic Information Flow Analysis**
POST@ETAPS 2014	**Information Flow Control in WebKit's JavaScript Bytecode**
ICSE 2014 Companion	**Exception handling for dynamic information flow control**
ICA3PP 2011	**Security Architecture for Virtual Machines**

Honors and Awards

2025	Invited lectures on Web Security at ICAN Faculty Development Program at IIT Madras
2024	Google India Research Award
2024	Google exploreCSR Award (along with Dr. Nipun Batra and Dr. Neeldhara Misra)
2024	Best Paper Award at BuildSec'24
2022	Google exploreCSR Award
2021	DST-Inspire Faculty Fellowship
2015	RS3 Best Paper Award for the CSF 2015 paper
2014	Second prize at the ACM SIGSOFT Student Research Competition at ICSE 2014
2011	Goyal Cash Prize for highest CGPA in CS from IIT Roorkee
2009 – 2011	MHRD Fellowship
2009	Best Performer in Training during internship at Infosys Technologies Ltd.

Service

Associate Editor
JCS 2025-Present
Journal of Computer Security

Program Co-Chair
PLAS 2025
20th Workshop on Programming Languages and Analysis for Security (PLAS 2025)

Track Co-Chair
ACM AsiaCCS 2026
Track on Software Security, Formal Methods and Programming Languages

Program Committee Member
USENIX Security Symposium – 2027
ACM Conference on Computer and Communications Security (CCS) – 2026
IEEE Computer Security Foundations Symposium (CSF) – 2022 - 25, 2027
The ACM Web Conference (WWW) – 2025
ACM ASIA Conference on Computer and Communications Security (AsiaCCS) – 2025
International Conference on Computer Aided Verification (CAV) – 2023
IEEE European Symposium on Security and Privacy (EuroS&P) – 2023 - 24
International Conference on Information and Systems Security (ICISS) – 2023 - 24

Grants and Funded Projects

2026 **Towards a Private yet Functional Web: Designing a Usable and Privacy-compliant Consent Management Framework for Online Browsing**
Lead-PI, Funded by ANRF ARG
INR 27,91,624 (IIT Gandhinagar budget)

2024 **Bridging Digital Divides Through Computing**
co-PI, Funded by Google exploreCSR Award 2024
INR 63,75,000

2024 **Towards a Secure and Private Web Browsing Experience**
PI, Funded by Google India Research Award 2024
INR 17,00,000

2023-2025 **Understanding and Ensuring the Privacy of User Data**
PI, Funded by SERB
INR 14,65,370

2023-2026 **Verified Security Properties of Group Messaging Protocols**
PI, Funded by CEFIPRA under DST-Inria grant
INR 36,13,292 (Indian-side budget)

2022 **Google exploreCSR Award**
PI, Funded by Google
INR 3,70,900

2021-2026 **Practical Dynamic Information Flow Control**
PI, Funded by DST as part of INSPIRE Faculty Award
INR 35,00,000